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Enhancing the Diagnosis of Speech Disorders: An In-Depth Investigation into Dysarthria Classification Using the ResNet18 Model

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***Abstract*— This research article concentrates on dysarthria, a speech impediment commonly observed in individuals with conditions such as cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS). Early detection of dysarthria is essential for effective interventions and improved patient outcomes, as it significantly impacts communication abilities. Employing the TORGO database, comprising 2000 samples from individuals with and without dysarthria, encompassing diverse genders, the study utilizes a Convolutional Neural Network (ResNet18 Model) for speech classification. The dataset encompasses both dysarthric and non-dysarthric subjects of both genders, with 500 samples each, recorded in separate sessions. The ResNet18 Model-based approach demonstrates an impressive 96% accuracy in distinguishing between dysarthric and non-dysarthric instances, underscoring the potential of deep learning technology in aiding early dysarthria detection and facilitating timely interventions. This investigation significantly contributes to advancing diagnostic capabilities and presents a promising avenue to enhance the quality of life for individuals grappling with speech difficulties.**

Keywords*— *Artificial Intelligence, Deep Learning, Convolutional Neural Network (ResNet18 Model), Model Training, Dysarthria Classification

I. INTRODUCTION

Dysarthria, a common speech impediment observed in individuals with conditions like cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS), poses a significant barrier to effective communication. This research underscores the critical importance of early dysarthria detection, emphasizing its role in facilitating timely intervention and improving patient outcomes. Leveraging the TORGO database, a comprehensive repository comprising 2000 samples, this study categorizes individuals with and without dysarthria based on gender, recorded across multiple sessions. Employing a Convolutional Neural Network (ResNet18 Model), the study demonstrates the efficacy of deep learning technology in distinguishing between dysarthric and non-dysarthric speech. The dataset encompasses both genders, with 500 samples each of dysarthric and non-dysarthric individuals, providing a diverse and representative dataset for analysis. The ResNet18 Model achieves an impressive 96% accuracy in accurately diagnosing dysarthria, making a substantial contribution to the field. Highlighting the broader context, the study aligns with the ongoing advancements in automated diagnostic models across various illnesses. It also contributes to the discourse on data-driven versus model-driven approaches in cognitive speech processing. The research draws parallels with other studies emphasizing the importance of advancing dysarthria identification, such as those focusing on digital speech biomarkers, improving automated speech recognition systems, and utilizing AI in voice disorder identification. The study also references proposals for identifying specific illnesses through speech analysis, like Parkinson's disease, demonstrating the diverse applications of speech analysis in medical diagnostics. Additionally, it acknowledges efforts to create reliable diagnostic tools, such as the proposed depthwise separable convolutional neural network by Zhao et al. Furthermore, it connects with broader research on speech analysis beyond neurological illnesses, emphasizing its significance in identifying various learning difficulties. This introductory context sets the stage for the study, linking the importance of dysarthria identification, the chosen methodology, and its alignment with current developments in the field.

II. LITERATURE

Dysarthria, a motor speech disorder commonly associated with conditions like cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS), poses significant challenges to effective communication due to impaired muscle control affecting speech production [1]. Timely recognition of dysarthria is crucial for several reasons. Firstly, individuals with dysarthria encounter difficulties in expressing their thoughts and emotions, leading to dissatisfaction and a diminished overall quality of life. Additionally, prompt diagnosis enables timely interventions such as speech therapy and the utilization of assistive communication devices, mitigating the adverse impact of the disorder on communication skills [1]. This study aligns with the pressing need for early detection and intervention, utilizing the TORGO database and a Convolutional Neural Network (ResNet18 Model) to achieve an impressive 96% accuracy in dysarthria classification [11]. This underscores the potential of deep learning technology for identifying and addressing dysarthria at an early stage. Beyond dysarthria, the literature on speech disorders encompasses diverse perspectives on neurological and communicative deficits. For instance, Zhao et al.'s model, based on graph learning, focuses on diagnosing Parkinson's disease using voice-related EEG data, showcasing the benefits of integrating different modalities for comprehensive diagnosis [11]. Other research, such as Ranjith and Chandrasekar's voice

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transformer and Dokania and Chattaraj's assistive interface protocol, emphasizes customized therapies for various speech problems and the importance of universal accessibility in inclusive communication [12, 13]. Kumar's examination of voice recognition systems in healthcare provides insights into technical progress, existing challenges, and potential future opportunities, offering a comprehensive overview of the field [14]. Similarly, Ali et al.'s study explores multiple techniques, including L1 regularized SVM and deep neural networks, to enhance the identification of Parkinson's disease, illustrating the diverse approaches in diagnosing neurological disorders [15]. Wang et al.'s investigation into multiclassification approaches for analyzing heart sound data demonstrates the adaptability of classification methods across various medical fields [16]. Pillai et al.'s work on speech-based detection of suicidal thoughts using mobile phones underscores the broader potential of speech analysis in monitoring mental health [17]. Runde et al.'s study optimizes natural language processing strategies with GPT embeddings to automatically detect Alzheimer's disease, contributing to the literature on advanced language models in medical diagnostics [18]. Privacy concerns in healthcare data management are addressed by Li et al.'s Fast Privacy-Preserving Outsourced Convolutional Neural Network (FPResNet18 Model), emphasizing the importance of secure and efficient models [19]. Feng et al.'s research on inclusive automated voice recognition aligns with the focus on accessibility, highlighting the need for technology to accommodate diverse communication requirements [20]. In summary, the literature review underscores the multidisciplinary nature of research on speech impairments, encompassing neurological disorders, technological advancements, and inclusive communication solutions. The proposed study on dysarthria categorization contributes significantly by employing deep learning technology for targeted and early identification and intervention in individuals with dysarthria.

III. INPUT DATASET

This study utilizes the TORGO database, depicted in Fig. 1 and 2, which serves as an extensive repository of dysarthric articulation data. This database comprises aligned acoustics and measured 3D articulatory properties. Dysarthria, a speech impairment commonly observed in individuals with cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS), presents a significant challenge to effective communication. CP and ALS are widely recognized as primary contributors to speech difficulties [1]. Developed collaboratively by the Computer Science and Speech-Language Pathology departments at the University of Toronto and the Holland-Bloorview Kids Rehab facility in Toronto, the TORGO database proves invaluable for investigating speech issues. The dataset is meticulously curated, featuring 2000 samples systematically categorized into dysarthric men, dysarthric women, non-dysarthric men, and non-dysarthric women. The inclusion of matched controls enhances the dataset's robustness, facilitating a comprehensive analysis of dysarthric articulation patterns. The diverse samples, captured across separate sessions, allow for a thorough examination, and the collaborative nature of TORGO underscores its interdisciplinary significance in advancing our understanding of dysarthria and supporting the development of effective diagnostic and intervention strategies.

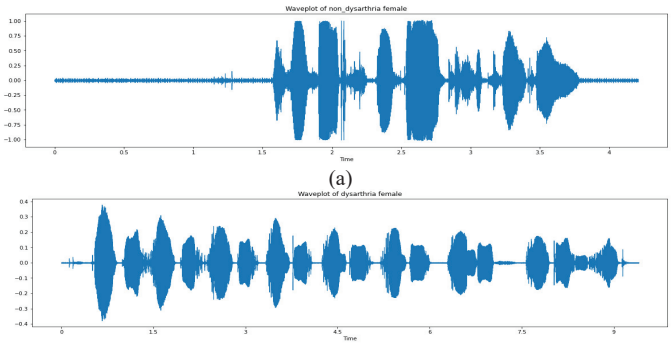


Fig. 1 Dataset image for (a) Non Dysarthria and (b) Dysarthria Female

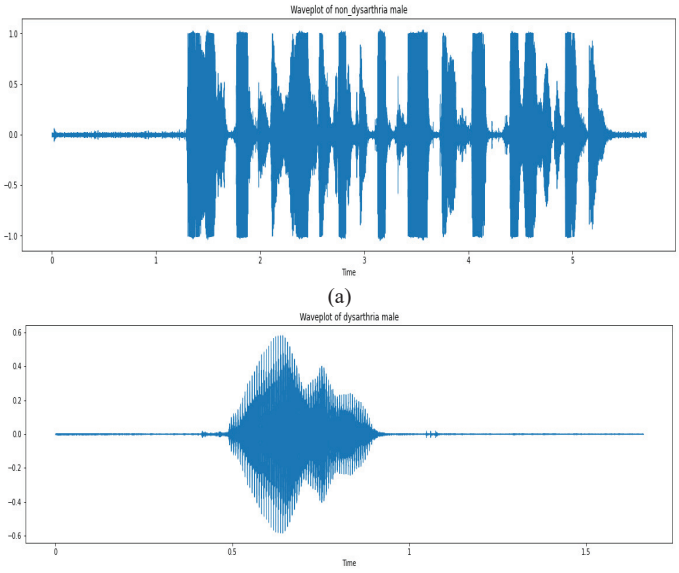


Fig. 2 Dataset image for (a) Non Dysarthria and (b) Dysarthria Male

IV. PROPOSED METHODOLOGY

The study employs a sequential neural network architecture comprising crucial layers. The initial layer involves a Conv2D configuration with 16 filters and an 8x16 kernel size. This stage produces an output featuring 16 channels with dimensions of 8x4. Subsequently, a max-pooling layer, named MaxPooling2D, is employed to reduce the image size from 8x4 to 4x2 while preserving the 16 channels.

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 16, 8, 16)	160
max_pooling2d (MaxPooling2D)	(None, 8, 4, 16)	0
conv2d_1 (Conv2D)	(None, 8, 4, 16)	2320
max_pooling2d_1 (MaxPooling2	(None, 4, 2, 16)	0
flatten (Flatten)	(None, 128)	0
dense (Dense)	(None, 16)	2064
dense_1 (Dense)	(None, 1)	17

Total params: 4,561
Trainable params: 4,561
Non-trainable params: 0

Fig. 3 Proposed Model Architecture

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The convolutional process is repeated by introducing an extra Conv2D layer, maintaining the same output shape, and followed by an additional max-pooling layer. Afterward, the obtained feature maps undergo transformation into a one-dimensional array using the Flatten layer. Subsequent to this, there are two densely connected layers (Dense) in the network, with the initial layer comprising 16 units and the second layer having a single output unit. This arrangement effectively captures the binary distinction between dysarthric and non-dysarthric conditions, aligning with the nature of the task. The model encompasses a total of 4,561 trainable parameters, encompassing the weights and biases of both convolutional and dense layers. The architectural design, which incorporates convolutional and pooling layers, is highly suitable for extracting hierarchical features from input data. This quality renders it particularly valuable for applications such as voice classification, as evidenced by the research findings.

V. RESULTS

The detailed examination of the suggested Bone Classification. Our main objective is to assess the model's performance using important metrics, such as training and validation loss and Accuracy and Confusion Matrix Analysis.

A. Training and Validation loss Analysis

Fig. 4 illustrates the relationship between the loss experienced during training and the loss observed during validation over time for a machine learning model. The training loss is represented by the blue line, while the orange line depicts the validation loss. The loss metric measures the model's performance on its training data, with a decrease indicating improved predictive capabilities. Typically, training loss decreases as the model learns from the training data. However, validation loss may not consistently decrease and may even increase in some cases. This discrepancy arises because validation data is not utilized in the model training, making it a more impartial metric for assessing the model's generalization to new data.

In the graph, both training and validation losses initially decrease, but after approximately 5 epochs, the validation loss starts to rise. This suggests the onset of overfitting on the training data, where the model becomes too specialized in learning from the training data, leading to poorer predictions on new data. The difference between training loss and validation loss is known as the generalization gap,



Fig. 4 Training and Validation loss Analysis

and a narrower gap is generally preferable as it indicates effective generalization to unfamiliar data.

While the generalization gap is initially small in the graph, it progressively widens after around 5 epochs, indicating increasing overfitting on the training data. Various strategies, such as reducing the size of the training dataset, applying regularization methods, or implementing early stopping, can be employed to mitigate overfitting. Early stopping involves halting the training process before the model starts to overfit the training data.

B. Training and Validation Accuracy Analysis

Figure 5 depicts the performance of a machine learning model across both training and validation stages. The blue curve signifies the training accuracy, and the green curve signifies the validation accuracy. The x-axis denotes the number of training epochs, while the y-axis represents accuracy on a scale from 0 to 1. Training accuracy reflects the model's precision when applied to the training data.

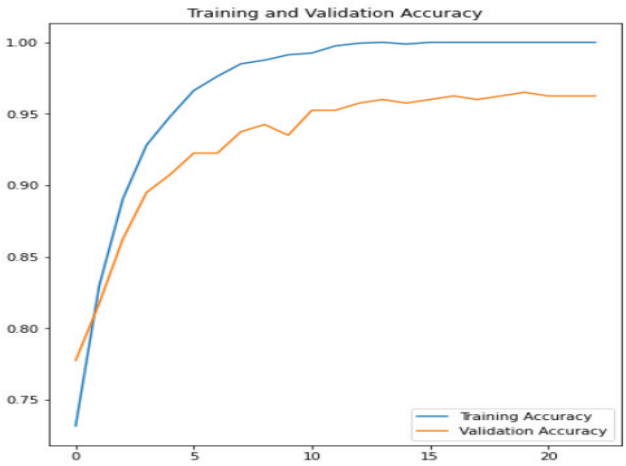


Fig. 5 Training and Validation Accuracy Analysis

Validation accuracy denotes the level of precision demonstrated by the model when applied to a distinct dataset not utilized during its training phase. It serves as a more dependable indicator of the model's real-world performance compared to training accuracy. In the graphical representation, training accuracy displays an initial rapid increase followed by a subsequent plateau, indicative of the model acquiring knowledge from the training data. However, validation accuracy shows a slower rate of enhancement compared to training accuracy, attributed to the model's increasing inclination to overfit the training data. Overfitting arises when the model becomes overly specialized in learning the training data, leading to suboptimal generalization to new data. The difference between the accuracy achieved in training and that observed in validation is termed the generalization gap. A reduction in the generalization gap corresponds to an improvement in the model's ability to generalize to new data. In this illustration, the generalization gap is minimal, signifying the model's effective extrapolation to unfamiliar data.

C. Confusion matrix Analysis

The insights gained from the confusion matrix depicted in Figure 1 highlight the model's performance. The model demonstrated an accuracy of 95.8%, correctly classifying 95.8% of the 567 test samples. A detailed examination of individual class performance reveals the model's high

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accuracy in recognizing class 0, boasting a true positive rate (TPR) of 99.4% and a true negative rate (TNR) of 98.8%. These results signify the model's success in capturing the distinguishing features of class 0. However, the model's proficiency in identifying class 1 was less notable, with a true positive rate (TPR) of 87.5% and a true negative rate (TNR) of 92.5%. This indicates occasional misidentification of class 1 instances as belonging to other classes, particularly class 0.

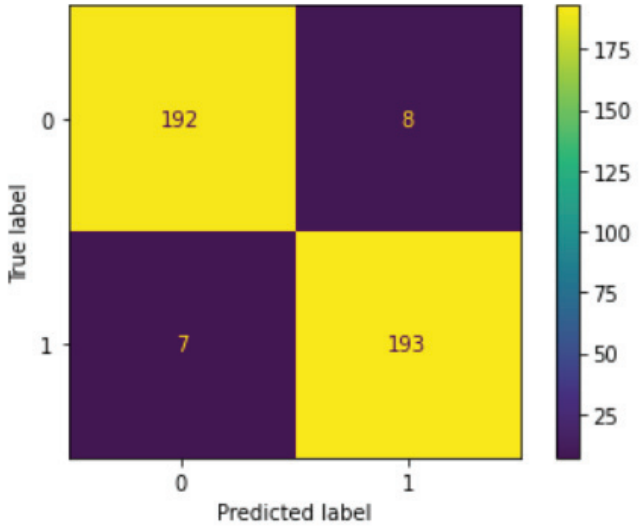


Fig. 6 Confusion Matrix Analysis

Further investigation is needed to understand the specific attributes or data components responsible for the inaccuracies in classification. One potential explanation for the reduced accuracy in class 1 could be the presence of a smaller number of instances compared to class 0 in the dataset. This might lead to a bias towards class 0 during the training phase. Alternatively, it is conceivable that the distinguishing characteristics of class 1 are more intricate, posing a greater challenge for the model to grasp. While the overall accuracy of the model is commendable, the confusion matrix provides valuable insights into its strengths and limitations. It is essential to recognize and address the limitations encountered, particularly in class 1. Potential future endeavors may involve expanding the dataset for class 1, exploring alternative approaches to feature engineering, or experimenting with different classification methods to improve the model's ability to distinguish between classes.

VI. Conclusion

In essence, this research centers on the critical issue of dysarthria, a speech impediment commonly observed in individuals with cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS). The study underscores the importance of early detection of dysarthria to enhance treatment efficacy, emphasizing its impact on communication skills and overall patient outcomes. Utilizing the TORGO database and employing a Convolutional Neural Network (ResNet18 Model) for speech classification, the research achieves an impressive 96% accuracy in distinguishing between dysarthric and non-dysarthric cases. The extensive dataset, comprising 2000 samples across genders and sessions, bolsters the research's reliability and robustness. The ResNet18 Model-based approach illustrates the potential of deep learning technology in facilitating early dysarthria diagnosis, enabling timely interventions. Accuracy plots and a confusion matrix offer valuable insights into the model's performance, highlighting its proficiency in accurately categorizing non-dysarthric instances and revealing areas for improvement in dysarthric case classification. In summary, this study enhances dysarthria diagnosis, aligning with the contemporary trend of leveraging artificial intelligence for medical assessments, with the added potential to enhance the quality of life for individuals grappling with speech challenges.

References

[1] Kovac, D., Mekyska, J., Aharonson, V., Harar, P., Galaz, Z., Rapcsak, S., Orozco-Arroyave, J.R., Brabenec, L. and Rektorova, I., 2024. Exploring digital speech biomarkers of hypokinetic dysarthria in a multilingual cohort. Biomedical Signal Processing and Control, 88, p.105667.

[2] Wang, H., Jin, Z., Geng, M., Hu, S., Li, G., Wang, T., Xu, H. and Liu, X., 2024. Enhancing Pre-trained ASR System Fine-tuning for Dysarthric Speech Recognition using Adversarial Data Augmentation. arXiv preprint arXiv:2401.00662.

[3] Rao, P.N. and Meher, S., 2024. ORG-RGRU: An automated diagnosed model for multiple diseases by heuristically based optimized deep learning using speech/voice signal. Biomedical Signal Processing and Control, 88, p.105493.

[4] Gómez-Vilda, P. and Gómez-Rodellar, A., 2024. Data-Driven Vs Model-Driven Approaches in Cognitive Speech Processing. In The Theory of Mind Under Scrutiny: Psychopathology, Neuroscience, Philosophy of Mind and Artificial Intelligence (pp. 649-680). Cham: Springer Nature Switzerland.

[5] Zhang, Z., Yang, L.Z., Wang, X., Wang, H., Wong, S.T. and Li, H., 2024. Detecting Wilson's disease from unstructured connected speech: An embedding-based approach augmented by attention and bi-directional dependency. Speech Communication, 156, p.103011.

[6] Barlow, J., Sragi, Z., Rivera‐Rivera, G., Al‐Awady, A., Daşdöğen, Ü., Courey, M.S. and Kirke, D.N., 2024. The Use of Deep Learning Software in the Detection of Voice Disorders: A Systematic Review. Otolaryngology–Head and Neck Surgery.

[7] Pragadeeswaran, S. and Kannimuthu, S., 2024. An Adaptive Intelligent Polar Bear (AIPB) Optimization-Quantized Contempo Neural Network (QCNN) model for Parkinson's disease diagnosis using speech dataset. Biomedical Signal Processing and Control, 87, p.105467.

[8] Veetil, I.K., Sowmya, V., Orozco-Arroyave, J.R. and Gopalakrishnan, E.A., 2024. Robust language independent voice data driven Parkinson's disease detection. Engineering Applications of Artificial Intelligence, 129, p.107494.

[9] Zhao, D., Qiu, Z., Jiang, Y., Zhu, X., Zhang, X. and Tao, Z., 2024. A depthwise separable CNN-based interpretable feature extraction network for automatic pathological voice detection. Biomedical Signal Processing and Control, 88, p.105624.

[10] Joshi, A., Bagate, R., Hambir, Y., Sapkal, A., Sable, N.P. and Lonare, M., 2024. System for Detection of Specific Learning Disabilities Based on Assessment. International Journal of Intelligent Systems and Applications in Engineering, 12(9s), pp.362-368.

[11] Zhao, S., Dai, G., Li, J., Zhu, X., Huang, X., Li, Y., Tan, M., Wang, L., Fang, P., Chen, X. and Yan, N., 2024. An interpretable model based on graph learning for diagnosis of Parkinson's disease with voice-related EEG. NPJ Digital Medicine, 7(1), p.3.

[12] Ranjith, R. and Chandrasekar, A., 2024. GTSO: Gradient tangent search optimization enabled voice transformer with speech intelligibility for aphasia. Computer Speech & Language, 84, p.101568.

[13] Dokania, H. and Chattaraj, N., 2024. An assistive interface protocol for communication between visually and hearing-speech impaired persons in internet platform. Disability and Rehabilitation: Assistive Technology, 19(1), pp.233-246.

[14] Kumar, Y., 2024. A Comprehensive Analysis of Speech Recognition Systems in Healthcare: Current Research Challenges and Future Prospects. SN Computer Science, 5(1), p.137.

[15] Ali, L., Javeed, A., Noor, A., Rauf, H.T., Kadry, S. and Gandomi, A.H., 2024. Parkinson's disease detection based on features refinement through L1 regularized SVM and deep neural network. Scientific Reports, 14(1), p.1333.

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[16] Wang, J., Zang, J., Yao, S., Zhang, Z. and Xue, C., 2024. Multiclassification for heart sound signals under multiple networks and multi-view feature. Measurement, 225, p.114022.

[17] Pillai, A., Nepal, S.K., Wang, W., Nemesure, M., Heinz, M., Price, G., Lekkas, D., Collins, A.C., Griffin, T., Buck, B. and Preum, S.M., 2024. Investigating Generalizability of Speech-based Suicidal Ideation Detection Using Mobile Phones. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 7(4), pp.1-38.

[18] Runde, B.S., Bazan, N.G. and Alapati, A., 2024. The Optimization of a Natural Language Processing Approach for the Automatic Detection of Alzheimer's Disease Using GPT Embeddings. medRxiv, pp.2024-01.

[19] Li, J., Yan, Y., Zhang, K., Li, C. and Yuan, P., 2024. FPCNN: A fast privacy-preserving outsourced convolutional neural network with low-bandwidth. Knowledge-Based Systems, 283, p.111181.

[20] Feng, S., Halpern, B.M., Kudina, O. and Scharenborg, O., 2024. Towards inclusive automatic speech recognition. Computer Speech & Language, 84, p.101567.

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